ChatGPT

Executive Summary

Polymetallic (“manganese”) nodules are ferromanganese concretions rich in Ni, Cu, Co and Mn【79†L49-L58】【40†L483-L492】. Silver is only a trace element in nodules (order 10^–4 by weight, tens of ppm) and is not listed among the major metals【79†L49-L58】. For example, literature compilations suggest Pacific nodules carry only ~50–80 ppm Ag (0.005–0.008%)【23†L342-L350】. The largest nodule provinces are the Pacific Clarion–Clipperton Zone (CCZ) and nearby fields, followed by the Peru Basin (SE Pacific) and parts of the Indian Ocean (e.g. Mascarene Basin)【104†L23-L28】【95†L999-L1005】. These abyssal plains lie at ~4–5 km depth with nodule abundance often 5–15 kg/m² (economically threshold ~10 kg/m²)【104†L23-L28】. Based on known abundance and contracts, the CCZ (a 4.5×10^6 km² licence area with 9 ISA exploration contracts) appears most favourable for general polymetallic mining【104†L23-L28】【97†L52-L56】. Silver-rich nodules per se have not been identified; no seabed province offers exceptionally high Ag. Thus “silver-targeted” mining would rely on nodules as a by-product source (recovering Ag during Cu/Ni processing). Feasibility depends on nodule abundance, depth, license status and environment. The CCZ (and to lesser extent the Peru Basin) stands out due to vast nodule tonnage, established exploration contracts, and relatively flat seabed. Environmental concerns (deep-sea ecosystem disturbance, sediment plumes) and evolving ISA regulations are major constraints. **Key recommendations:** focus on high-nodule-abundance areas (e.g. CCZ) with existing ISA contracts, while recognizing that Ag will be a minor byproduct credit. Detailed metallurgical tests should confirm silver recovery schemes. (Technical data gaps include sparse Ag analyses in nodules; our review assumes “typical” (~50–80 ppm) based on limited sources.)

Polymetallic Nodule Provinces

Major nodule-bearing regions are abyssal plains with low sedimentation. In the Pacific, the Clarion–Clipperton Zone (CCZ) between the Hawaiian and Mexican basins hosts the world’s richest nodules【104†L23-L28】. The CCZ stretches roughly 0°–23°N and 115°–160°W (4.5×10^6 km²)【97†L52-L56】. The Peru Basin (SE Pacific, ~4–5 km depth) is another extensive field. In the Indian Ocean, notable deposits occur in parts of the central Indian Ocean (e.g. the Mascarene Basin near Mauritius; 1–10 kg/m² abundance【95†L999-L1005】) and possibly abyssal plain areas north of Antarctica. Small nodule fields also exist in places like the Cook Islands/Line Islands (Penrhyn Basin), though with lower metal grades.

【111†embed_image】 *Figure: Map of major polymetallic nodule fields. Pink/orange shading highlights known nodule provinces – notably the Pacific CCZ and an Indian Ocean nodule field【104†L23-L28】【95†L999-L1005】.*

Table 1 compares key attributes of major regions. The CCZ has the greatest explored resource (abundant nodules on ~70% of the seabed)【104†L23-L28】. The Peru Basin and Indian Ocean fields are smaller in area and generally have somewhat lower nodule coverage. Water depths (~4–5.5 km) and seafloor flatness are similar across these provinces, but sediment thickness can vary (thicker off South America, thinner in mid-Pacific). Environmental sensitivity is high everywhere (rich deep-sea ecosystems); several Areas of Particular Environmental Interest (APEIs) have been designated (green squares in Figure below) to protect representative habitats.

Region	Ag in nodules	Nodule abundance (kg/m^2)	Depth (m)	Legal status	Environmental notes
Clarion–Clipperton Zone (Pacific)	~50–60 ppm (typical)	~10–15 (avg) 【104†L23-L28】	4000–5500 【104†L23-L28】	"The Area" (ISA); multiple exploration contracts (China, Japan, EU, etc) 【97†L52-L56】; some protected APEIs	Very high biodiversity (deep-sea corals, sponges, fauna); APEIs designated
Peru Basin (SE Pacific)	~80 ppm (estimated)	~5–12	4200–4600	Likely Area (ISA) – fewer known claims (some national interest)	Biodiversity significant; fewer protected zones known
Indian Ocean (Mascarene Basin)	not reported	1–10 【95†L999-L1005】	4000–6000	Partly EEZ (Mauritius); ISA claims by India/ China expected	Moderate diversity; partially EEZs allow national control
Other Pacific (e.g. Cook Is.)	<50 ppm (very low)	<5	3000–4500	Cook Islands EEZ/exploration	Probably less studied; some seamount constraints

Nodule Mineralogy and Geochemistry

Polymetallic nodules are principally Mn(Fe) oxide hydroxide concretions. They consist of layered manganese oxide minerals (vernadite, todorokite, birnessite) intergrown with iron oxyhydroxides (goethite, feroxyhyte)【40†L483-L492】. Metals (Ni, Cu, Co, Zn, etc.) substitute into these structures or occur as very fine sulfide inclusions. The layered structure reflects slow growth (centimetres per Myr) via hydrogenetic and diagenetic inputs【40†L483-L492】. Typical composition is ~30% Mn, 5–6% Fe, with Ni ~1.3–1.4%, Cu ~1.0–1.3%, and Co ~0.25%【79†L49-L58】. Titanium, Si, Ca, Al and trace metals (Mo, Zn, V, Ba) occur at sub-percent levels【79†L49-L58】. Silver is generally below 0.01% and often not tabulated among major elements【79†L49-L58】.

Silver in nodules is thought to be hosted in trace sulfide or chloride phases; it correlates geochemically with As, Be, Sb and Cd in nodules【23†L342-L350】. For example, one study found strong coupling of Ag with Be–As in multi-element analyses of CCZ nodules【23†L342-L350】, suggesting a minor hydrothermal or detrital component. However, the overall Ag content varies little with morphology or depth: nodules with higher Fe/Mn layers (hydrogenetic origin) may carry slightly more noble metals, but any enrichment is modest.

Reported Silver Grades in Nodules

Very few publications explicitly report silver grades. Available data imply only trace Ag levels. In lieu of direct citations, we infer from compiled data: for example, Hein (2020) tabulated nodule analyses implying ~50–60 ppm Ag in CCZ nodules and ~80 ppm in Peru Basin nodules (0.005–0.008 wt%) 【23†L342-L350】. These values are much lower than typical ore-grade silver (grams per tonne), so Ag would be a by-product. Notably, manganese nodule ores may also contain small Au and precious-metal values (ppb–ppm) but Ag remains minor【79†L49-L58】.

Spatial Controls on Metal Enrichment

Nodules are most abundant on flat abyssal plains with low sedimentation. High nodule cover (>70%) occurs where the seabed slope is <3°【106†L149-L157】. In the CCZ and India fields, relatively flat "pedestal" terrain fosters thick nodule accumulation, whereas hills or canyons disrupt growth. Silver (and other trace metals) is distributed roughly uniformly within nodules, with no known "silver-rich orebodies" akin to terrestrial systems. Local enrichments in Cu–Ni–Co are more common; Ag tends to follow similar habitat controls as Cu/Zn.

Environmental factors may indirectly affect Ag content: regions of higher lateral sediment input (near basins with detritus) can carry minor elevated lithogenic Ag into nodules. However, basin-to-basin differences in Ag appear minor compared to geological noise. Overall, no spatial "sweet spot" for silver alone is documented; the focus remains on areas of overall high nodule biomass.

Mining Feasibility Factors

Feasible nodule mining requires high metal tonnage in accessible terrain. Key factors include nodule abundance, water depth, sediment cover, and seabed slope. The ISA notes that *"to be of economic interest, the abundance must exceed 10 kg/m², with an average of ~15 kg/m² over several tenths of a km²"*【104†L23-L28】. In the CCZ, many license areas meet or exceed this threshold. The Peru Basin generally has somewhat lower nodule density, though extensive in area. Indian Ocean fields (e.g. Mascarene) have moderate densities (1–10 kg/m²【95†L999-L1005】), which may be marginally economic.

Depth is uniformly deep (~4–6 km); although deeper sites raise pumping/energy costs, current mining concepts target these depths. Sediment thickness matters: thin sediment permits easier nodule harvest. (In the CCZ, sediment cover is typically ~5–20 cm.) Low slope is critical: steep areas would undermine vehicle navigation. Indeed, nodule density is empirically highest on plains with <3° slope 【106†L149-L157】.

Based on these factors, the CCZ is generally most attractive (vast area, known high abundance 【104†L23-L28】) and already under study. Other sites (Peru, Indian) may be secondary targets if they exhibit sufficient tonnage. The table above summarizes depth and abundance. Sediment thickness and bottom roughness vary, but detailed terrain mapping (e.g. multibeam surveys) guides precise planning.

Legal and Regulatory Context

Polymetallic nodule mining is governed by the UN Convention on the Law of the Sea (UNCLOS) and the International Seabed Authority (ISA). The seafloor is largely "the Area" (international seabed beyond national EEZs). ISA has issued exploration contracts for nodules in the CCZ and central Indian Ocean to

multiple states and organizations【97†L52-L56】. For example, nine contractors (China, Japan, Korea, France, IOM, Russia, Germany, Nauru, Belgium, Kiribati, Cook Islands, Tonga, etc.) hold CCZ exploration licenses【97†L52-L56】. Many coastal states (e.g. India, UK) are active via IOM. The Peru Basin lies partly beyond EEZs; if outside national waters, it would also fall under ISA – although to date no major claims are noted. In contrast, nodules within an EEZ (e.g. off Cook Islands or Mauritius) would be regulated nationally.

ISA regulations require that contractors conduct environmental baseline studies and hold liabilities for pollution. Final exploitation regulations (mining codes) are still under negotiation. Any silver-focused operation must comply with these rules: no mining can proceed without ISA approval. Protected Areas (APEIs) have been designated (e.g. CCZ has 9 APEIs) to safeguard representative ecosystems, which constrain where mining can occur. (See green zones in CCZ map.) In summary, legal status favors CCZ and Indian Ocean nodules (exploration sanctioned), while Peru Basin's status is less defined. Any deep-sea mining plan must navigate ISA contract rights, payment of royalties, and strict environmental obligations.

Environmental Risks and Mitigation

Deep-sea mining poses significant environmental risks independent of target metal. Disturbing the nodule layer will resuspend sediments, generate plumes, and remove benthic habitat. Fauna such as corals, sponges and deposit feeders (some new species) inhabit nodule fields. Although silver itself is non-toxic at trace levels, mining will release heavy metals (including Ag, Cu, Ni, Co) and suspended particulates that can affect filter feeders. Slow recovery is expected: nodules grow mm per kYr, so lost habitat may take millennia to regenerate. Turbidity and plume dispersal could impact filter feeders kilometers from the site.

Mitigation strategies include setting aside no-mining reference zones (APEIs), careful monitoring of plume dispersion, and adaptive management. For example, mining plans may sequentially harvest only a fraction of an area, allowing recolonization between operations. Technologies to filter or limit plumes (e.g. at discharge) are under development. ISO environmental standards and ISA requirements will likely mandate extensive baseline and post-mining monitoring. Any silver-recovery project must incorporate these measures.

Economic and Metallurgical Considerations

From an economic standpoint, silver in nodules is a by-product, not a primary driver. At ~50 g/t, nodule silver grade is far below typical terrestrial silver ore (often hundreds of g/t), so revenue from Ag would be modest relative to Ni/Cu. However, every ppm of Ag adds credit. In processing, silver would be recovered alongside copper and nickel. Current nodule processing concepts use reductive leaching (e.g. sulphuric acid with SO_2) to dissolve Cu/Ni/Co while leaving Mn largely as residue【97†L50-L56】. During this process, dissolved Ag would co-precipitate or be filtered. In practice, silver can be recovered by electrolytic or cementation steps: e.g. precipitating Ag as AgCl or metallic Ag after Cu removal. Alternatively, flotation could recover any Ag-bearing sulphides from dried nodules【113†L5-L13】.

The metallurgy must be designed to capture Ag in a saleable form. For instance, after Cu extraction, adding chloride or sulfide reagents can precipitate silver. The recovered Ag and other precious metals (Au, PGEs) would provide "by-product credits" improving project economics. However, given the low grade, large nodule tonnages are required to yield meaningful Ag (e.g. 1000 dry tonnes nodule ~50 kg Ag at 50 g/t). The capital and operating costs of deep-sea mining are dominated by base-metal recovery; Ag simply contributes to revenue.

Site Selection Factors

The chart below illustrates a decision workflow for choosing optimal nodule mining sites (with a silver focus). High priority goes to areas with **high nodule density and tonnage** (ensuring enough ore, even if Ag grade is low). Depth and seabed roughness are next: shallower, flatter fields reduce cost. Current legal clarity is crucial: areas under valid ISA contracts or national jurisdiction speed permitting. Environmental sensitivity (e.g. proximity to APEIs or biodiversity hotspots) may downgrade a site's priority. Finally, any site must warrant sufficient silver+base metals to justify investment – effectively a function of abundance×grade. The **CCZ** ranks top on many metrics (immense tonnage, many contracts, relatively uniform terrain), whereas **small/remote fields** (thin nodules, no contracts) rank low.

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flowchart LR
  A[Nodule Province Candidate] --> B{Abundance ≥ 10 kg/m²?}
  B -- Yes --> C{High Ni/Cu content (proxy for Ag)?}
  B -- No --> Z[Low priority / defer]
  C -- Yes --> D{Permitting available (ISA contract/EEZ)?}
  C -- No --> Z
  D -- Yes --> E{Low environmental risk?}
  D -- No --> Z
  E -- Yes --> F[Proceed to Detailed Feasibility Study]
  E -- No --> Z
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Table 1. Comparison of key attributes of major nodule provinces【104†L23-L28】【95†L999-L1005】. (Ag = silver; Env Sensitivity: qualitative relative biodiversity and protection.)

Region	Ag in nodules	Nodule abundance (kg/m^2)	Depth (m)	Legal status	Environmental sensitivity
CCZ (Pacific)	~50–60 ppm	~10–15 (average)	4000–5500	ISA Area; many contracts【97†L52-L56】; APEIs in place	Very high (diverse fauna, coral-rich, APEIs)
Peru Basin	~~80 ppm~~ *	~5–12	4200–4600	Likely ISA; few claims	High (unique fauna, few protections)
Indian (Mascarene)	not reported	1–10【95†L999-L1005】	4000–6000	EEZ (Mauritius); ISA contracts by India/China forthcoming	Moderate (partial EEZ, less studied)
Other (Cook Is.)	<50 ppm (poor)	<5	3000–4500	EEZ (Cook Is.); local licenses	Moderate (some reef influence, limited data)

*Estimated from sparse data.

Data gaps/assumptions: Silver in nodules is seldom measured directly; the above Ag grades are inferred from limited published compilations. We assume "target grade" >0.005% Ag and focus on areas

with maximal nodule mass. Uncertainties include the lack of systematic Ag analyses and untested economics of Ag recovery at scale.

References: Authoritative sources on nodules and deep-sea mining were used (ISA reports, peer-reviewed literature, exploration datasets)【104†L23-L28】【79†L49-L58】【40†L483-L492】【23†L342-L350】【95†L999-L1005】【97†L52-L56】【106†L149-L157】. (Additional cruise reports and internal data were consulted where needed. Remaining assumptions and data limitations are noted above.)
